

Project Title	IntegrityFlow An Autonomous AI Examination Monitoring and Secure Environment System
Supervisor Name	Mr. Abdullah
Co-Supervisor Name	
Group information	
Roll number	Student's Name
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Module's Name	Module Description with Deliverables
1. Secure Environment Enforcement	- Windows candidate desktop application using Electron. - Application whitelist enforcement with examiner-permitted tools and process monitoring. - Clipboard and display restrictions, USB-storage monitoring and pre-existing-file checks. - Explicit permitted-file exceptions and online/physical-lab exam modes.
2. AI Monitoring	- Local webcam processing using OpenCV and MediaPipe face landmarks when available. - Head-turn, missing-face, multiple-face/person and camera-obstruction signals. - YOLO26n ONNX inference for configured phone/book checks. - Calibration and confidence/persistence filters; alerts support examiner review.
3. Evidence Capture & Delivery	- Event-triggered desktop screenshots for system events and webcam frames for visual events. - Timestamped JPEG evidence linked to the candidate session and violation. - Local durable queues, stable event IDs and retries through Python and Electron. - Server evidence storage and retrieval, including valid delayed evidence after exam completion.

IntegrityFlow - System Modules

Module's Name	Module Description with Deliverables
4. Candidate Self-Check System	- Step-by-step camera/face calibration, microphone and reference-voice checks. - Application, USB-storage and display verification before exam entry. - Physical-lab mode bypasses camera and voice checks.
5. Violation Severity Scoring Engine	- Server-side severity weighting and time-decayed aggregation of violation records. - Per-candidate risk score and examiner review priority. - Review decisions affect scoring; risk is not a probability of cheating.
6. Examiner Dashboard & Exam Management	- Create/manage exams, release papers and configure rules and permitted applications. - Live Monitoring, Active Candidates, Priority Queue and exam-scoped evidence review. - Socket.IO updates restricted by examiner ownership/admin access. - Confirm/dismiss alerts, review submissions and inspect historical exams.
7. Answer Submission System	- Candidate paper viewing, typed answers and supported file uploads. - Local text drafts, exam timing and submission controls. - Answers linked to the candidate session; examiner retrieval and download.
8. Audit Report / Exam Summary	- Historical exam summaries, final risk scores and violation/risk distributions. - CSV export and browser print/Save PDF. - Detailed alert logs and screenshots remain in the separate evidence viewer; an all-evidence export is remaining work.

IntegrityFlow - Additional Implemented Modules

Module's Name	Module Description with Deliverables
9. Voice Monitoring & Speaker Comparison	- WebRTC VAD selects speech; Resemblyzer generates speaker-reference vectors. - Cosine-similarity comparison and repeated valid mismatches flag possible other-speaker activity. - Suspicious audio clips support review; consent discloses voice/evidence collection. - Offline labelled-recording evaluation supports testing; no speaker-count or accuracy guarantee.
10. Authentication & Administration	- Teacher/admin authentication, account verification/reset and role/ownership checks. - Administrative user/asset management and activity audit records. - Teacher information and consistent light/dark dashboard themes.
11. Examiner-Candidate Communication	- Exam chat, broadcasts and examiner warnings. - Session termination and time-extension controls. - Candidate status updates and re-verification requests during the exam.

Respected Evaluator, if you recommend an improvement or suggest a new module, please mention it in this section.

Evaluator's Name:

Suggestions

Supervisor Signature

Signature FYP Committee Member